

Topic B11

Gas exchange in humans

Concept guide · Worked application · Cambridge-style practice · Fully explained answers

What this guide teaches

A complete explanation of the syllabus concepts, biological mechanisms, common misconceptions, data interpretation and original exam-style practice for this topic.

Syllabus map

B11.1	Gas exchange system
B11.2	Alveoli
B11.3	Ventilation
B11.4	Effects of exercise and smoking

How to study: retrieve definitions, cover the application, then complete the practice paper without using the notes.

1. Gas exchange system

Air passes through trachea, bronchi and bronchioles to alveoli. Cartilage keeps larger airways open; mucus traps particles and cilia move it away. Gas exchange is diffusion of oxygen into blood and carbon dioxide out.

Examiner insight

Distinguish ventilation, movement of air, from gas exchange by diffusion.

2. Alveolar adaptations

Millions of alveoli provide large area. Their walls and capillary walls are one cell thick and moist. Dense blood flow and ventilation maintain steep concentration gradients. Red cells rapidly bind oxygen, helping preserve the gradient.

Examiner insight

Explain how each adaptation increases diffusion rate.

3. Ventilation

During inhalation, external intercostal muscles contract, ribs move up and out, diaphragm contracts and flattens, thoracic volume increases and pressure falls, so air enters. Exhalation reverses these changes at rest.

Examiner insight

Pressure falls because volume increases; air then moves down a pressure gradient.

4. Exercise and smoking

Exercise raises ventilation rate and depth to supply oxygen and remove carbon dioxide. Tar damages cilia and contains carcinogens; carbon monoxide reduces oxygen transport; smoke damages alveolar walls, reducing area and elasticity in emphysema.

Examiner insight

Connect smoking damage to the specific loss of transport or gas-exchange function.

Worked application

Situation: Alveolar walls are destroyed by smoking.

Prediction: Diffusion capacity and oxygen uptake decrease.

Reason: Surface area falls and ventilation becomes less effective due to loss of elasticity.

Exam technique

High-value habits

Define terms precisely, describe visible patterns before explaining them, link structure to function, use data values as evidence, identify variables explicitly and answer the command word directly.

Retrieval checklist

- Recall five key definitions without notes.
- Explain one process as a linked cause-and-effect chain.
- Interpret the data set and identify a limitation.
- Complete the timed practice and mark only after finishing.

Practice Paper

Topic B11

Gas exchange in humans

Instructions

Answer all questions. Select one option for multiple choice. Use values from tables where relevant. Write explanations on the lines. Individual parts carry 1-3 marks.

Candidate name:

Class / date:

Total marks: 50

Suggested time: 60 minutes.

Section A - Multiple choice

Choose the best answer for each question.

1. Which statement best defines ventilation? [1]
 - A movement of air into and out of the lungs
 - B diffusion of oxygen and carbon dioxide between alveoli and blood
 - C a microscopic air sac where gas exchange occurs
 - D a sheet of muscle below the lungs that changes thoracic volume
2. Which statement best defines gas exchange? [1]
 - A a microscopic air sac where gas exchange occurs
 - B diffusion of oxygen and carbon dioxide between alveoli and blood
 - C a sheet of muscle below the lungs that changes thoracic volume
 - D damage to alveolar walls that reduces surface area and elasticity
3. Which statement best defines alveolus? [1]
 - A a sheet of muscle below the lungs that changes thoracic volume
 - B damage to alveolar walls that reduces surface area and elasticity
 - C a microscopic air sac where gas exchange occurs
 - D movement of air into and out of the lungs
4. Which statement best defines diaphragm? [1]
 - A damage to alveolar walls that reduces surface area and elasticity
 - B movement of air into and out of the lungs
 - C diffusion of oxygen and carbon dioxide between alveoli and blood
 - D a sheet of muscle below the lungs that changes thoracic volume
5. Which statement best defines emphysema? [1]
 - A damage to alveolar walls that reduces surface area and elasticity
 - B movement of air into and out of the lungs
 - C diffusion of oxygen and carbon dioxide between alveoli and blood
 - D a microscopic air sac where gas exchange occurs
6. Which statement about gas exchange system is correct? [1]
 - A Diffusion of gases across alveoli.
 - B Air passes through trachea, bronchi and bronchioles to alveoli.
 - C Movement of ribs during breathing.
 - D A blood vessel carrying air.
7. Which statement about alveolar adaptations is correct? [1]
 - A Movement of ribs during breathing.
 - B A blood vessel carrying air.
 - C Millions of alveoli provide large area.
 - D A cartilage ring in the trachea.

8. Which statement about ventilation is correct? [1]

- A A blood vessel carrying air.
- B A cartilage ring in the trachea.
- C Infection of ciliated cells only.
- D During inhalation, external intercostal muscles contract, ribs move up and out, diaphragm contracts and flattens, thoracic volume increases and pressure falls, so air enters.

9. Which statement about exercise and smoking is correct? [1]

- A Exercise raises ventilation rate and depth to supply oxygen and remove carbon dioxide.
- B A cartilage ring in the trachea.
- C Infection of ciliated cells only.
- D Diffusion of gases across alveoli.

10. Which prediction is most appropriate when alveolar walls are destroyed by smoking? [1]

- A Diffusion of gases across alveoli.
- B A blood vessel carrying air.
- C Diffusion capacity and oxygen uptake decrease.
- D Infection of ciliated cells only.

Section B · Structured questions

11 Biologists use precise vocabulary when studying gas exchange in humans.

(a) Define ventilation.

[2]

(b) Define gas exchange.

[2]

(c) Distinguish alveolus from the incorrect idea: “a blood vessel carrying air”.

[2]

(d) State two important features of gas exchange system.

[2]

[Total: 8]

12 Alveolar walls are destroyed by smoking.

(a) State the most likely biological outcome.

[2]

(b) Explain the outcome using a linked biological mechanism.

[3]

(c) Predict what would happen if the relevant change were reversed. [2]

(d) State one assumption made in this prediction. [1]

[Total: 8]

13 A student records breaths per minute under three conditions.

condition	breaths per minute
rest	12
walking	20
running	34

(a) Calculate the difference between the highest and lowest recorded values. [2]

(b) Describe the pattern in the results. [2]

(c) Explain the pattern using biological knowledge. [3]

(d) State one limitation of drawing a conclusion from these data. [1]

[Total: 8]

14 A student plans to compare exhaled carbon dioxide before and after exercise.

(a) Identify the independent variable. [1]

(b) Identify the dependent variable. [1]

(c) State two control variables. [2]

(d) Describe how to obtain reliable results across a suitable range. [3]

(e) Suggest one specific improvement. [1]

[Total: 8]

15 Apply your knowledge of gas exchange in humans to an unfamiliar context.

(a) Define diaphragm. [2]

(b) Explain the biological importance of ventilation. [3]

(c) Correct this statement: "infection of ciliated cells only".

[2]

(d) State one further question that could be investigated.

[1]

[Total: 8]

Worked solutions

Remove or fold back this section while completing the practice paper.

Marking routine

Mark each part separately. Award points for accurate biology, use of evidence and linked reasoning, then diagnose the first error.

Solutions · Multiple choice

- 1 A. Movement of air into and out of the lungs.
- 2 B. Diffusion of oxygen and carbon dioxide between alveoli and blood.
- 3 C. A microscopic air sac where gas exchange occurs.
- 4 D. A sheet of muscle below the lungs that changes thoracic volume.
- 5 A. Damage to alveolar walls that reduces surface area and elasticity.
- 6 B. Distinguish ventilation, movement of air, from gas exchange by diffusion.
- 7 C. Explain how each adaptation increases diffusion rate.
- 8 D. Pressure falls because volume increases; air then moves down a pressure gradient.
- 9 A. Connect smoking damage to the specific loss of transport or gas-exchange function.
- 10 C. Surface area falls and ventilation becomes less effective due to loss of elasticity.

Solutions · Structured questions

11. (a) movement of air into and out of the lungs [2]. (b) diffusion of oxygen and carbon dioxide between alveoli and blood [2]. (c) a microscopic air sac where gas exchange occurs; not a blood vessel carrying air [2]. (d) Credit any two accurate features from the guide: Air passes through trachea, bronchi and bronchioles to alveoli. Cartilage keeps larger airways open; mucus traps particles and cilia move it away. Gas exchange is diffusion of oxygen into blood and carbon dioxide out. [2].

12. (a) Diffusion capacity and oxygen uptake decrease [2]. (b) Surface area falls and ventilation becomes less effective due to loss of elasticity [3]. (c) Accept the biologically justified opposite trend when the causal factor is reversed [2]. (d) One valid assumption, such as other relevant conditions remaining constant [1].

13. (a) $34 - 12 = 22$ [2]. (b) The values change from 12 to 34, with the highest value 34. [2]. (c) Greater respiration demand increases carbon dioxide stimulation of the breathing centre [3]. (d) Only three conditions / no repeats / small dataset / uncontrolled variation [1].

14. (a) Exercise state [1]. (b) Time for hydrogencarbonate indicator change [1]. (c) Same person, breath volume, indicator and temperature [2]. (d) Use several values across a sensible range, repeat each condition, identify anomalies and calculate means [3]. (e) Measure carbon dioxide with a sensor and repeat [1].

15. (a) a sheet of muscle below the lungs that changes thoracic volume [2]. (b) Award linked points from: During inhalation, external intercostal muscles contract, ribs move up and out, diaphragm contracts and flattens, thoracic volume increases and pressure falls, so air enters. Exhalation reverses these changes at rest. [3]. (c) Correct statement: emphysema means damage to alveolar walls that reduces surface area and elasticity [2]. (d) Any testable relevant biological question [1].

Self-assessment

Classify the first lost mark as definition, biological knowledge, causal link, data use, experimental design or communication.

Evidence used: Cambridge IGCSE Co-ordinated Sciences 0654 syllabus 2025-2027 and June 2026 papers, mark schemes and Principal Examiner Report supplied by the user. All practice questions are newly written.